STRATHMORE BUSINESS SCHOOL

MASTER OF BUSINESS ADMINISTRATION

END OF SEMESTER EXAMINATION

MBA 8104 QUANTITATIVE ANALYSIS FOR BUSINESS MANAGEMENT

Date: Thursday 26th July 2022 **Time: 3** Hours

Instructions: Answer **Question ONE** and any other **FOUR questions**

Question 1 (`20 Marks) COMPULSORY

(a) A financial analyst at Energy Regulatory Authority of Kenya would like to estimate the mean monthly electricity bill for a single-family house in December. Based on similar studies the standard deviation is estimated to be $20.00. A 99% level of confidence is desired, with an accuracy of ± $5.00. Determine how large a sample is required **(2 Marks)**

(b) The distribution of starting salaries of men who received Master of Business Administration (MBA) degrees last year had a mean equal to $15,000 and a standard deviation equal to $1,200. A random sample of 36 starting salaries of women who received MBA degrees last year had a sample mean equal to $15,600

(i) Determine the probability that a randomly selected sample of 36 stating salaries of men who graduated last year would have a sample mean at least as large as $ 15,600 **(3 Marks)**

(ii) Based on your findings in part a(i) above, does it appear that the starting salaries of women should also be characterized by the same probability distribution that characterizes men's starting salaries? Explain. **(2 Marks)**

(c) The descriptive statistics for weekly wages (in hundred Kshs) earned by company workers are given below:

Variable	Total Count	Mean	StDev	Minimum	Q1	Median	Q3
weekly wages	40	32.83	10.12	15.00	25.00	31.50	41.75

Variable	Maximum	Kurtosis
weekly wages	54.00	-0.77

(i) Develop the 95% and 99% confidence intervals for the mean weekly wages. **(4 Marks)**

(ii) Suppose the board of directors recommend salary increments for these workers. Which confidence level would you recommend when implementing this recommendation? Why?

(2 Marks)

(d) In a survey of graduate students, the following data were obtained on student's first reason for application to the school in which they were matriculated.

Enrolment status	Reasons for application		
	School quality (Q)	School cost or convenience (C)	Other (O)
Full time student (F)	754	62	61
Part time student (P)	157	7	6

Suppose one student is chosen at random, determine the probability that the student

(i) preferred **other** reasons for application. **(1 Mark)**

(ii) was part time given that the student preferred application due to school cost or convenience.

(3 Marks)

(e) Procedures to maintain comparable health workers' salaries between those employed by the government and their counterparts in the private hospitals are mandated by a given country's law. The design of the survey resembles a paired difference design because the occupation and experience of employees in the private and government hospitals are matched as closely as possible before a comparison. The summary *minitab* output below shows the annual salaries for 12 pairs of health workers in the sample, matched on job level and experience.

Descriptive Statistics: Private, Government hospitals

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Total
Variable   Count  Mean  StDev  Variance  Minimum     Q1  Median
Private       12  20967   8511  72438788    12500  14825   18350
Government    12  20304   8682  75378390    11750  14575   18150


Variable      Q3    Maximum   Range    Kurtosis
Private      23050    42100    29600      2.75
Government   21475    43200    31450      4.09
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Required:

(i) Giving reasons, comment on the nature of distribution of salaries for the two sectors **(1 Mark)**

(ii) Based on the results above, determine the coefficient of variation and comment on the results **(2 Marks)**

Question 2 (`10 Marks)

(a) Suppose Jane buys a stock whose return (including both dividends and the change in stock prices) depends on whether the nation's gross domestic product is rising, constant or falling. If the gross domestic product is rising, the return is 20 per cent per dollar invested; if its constant, the return is 5 per cent; and if its falling, the return is -10 (minus 10) per cent. Jane believes that it is equally likely that the gross domestic product will rise, remain constant, or fall. Determine

(i) Develop the probability distribution table **(1 Mark)**

(ii) Determine expected value of return from this stock **(2 Marks)**

(iii) Evaluate the standard deviation in percent of the return **(2 Marks)**

(b) One of the purposes of an audit is to detect the presence of material errors, procedural errors, or errors in judgement in the recording of accounting information. Suppose a firm has been retained to conduct an audit of the accounting practices of a firm in which accounts are processed by two divisions, a wholesale division and a retail division. The auditor knows that 70% of all Accounts are retail accounts. Furthermore, the auditor knows that 10% of all retail accounts and 20% of all wholesale accounts contain some type of accounting error. If the auditor observes an accounting error in an account, determine which division processed this accounting error. **(5 Marks)**

Question 3 (`10 Marks)

A financial Analyst wants to know whether profits in the year 2021 (Y) is related to it's profits in the year 2020 (X). The analyst obtains the following regression output regarding Y and X for 10 banking institutions, both X and Y are expressed in thousand dollars.

Regression Analysis: Y versus X

Predictor	Coef	SE Coef	T	P
Constant	1.691	1.863	0.91	0.394
X	0.3813	0.5385	0.71	0.502

S = 0.681116 R-Sq = 6.7% R-Sq(adj) = 0.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	0.2326	0.2326	0 .50	0.502

Obs	X	Y	Fit	SE Fit	Residual	St Resid
1	*	2.500	*	*	*	*
2	3.10	2.000	2.873	0.289	-0.873	-1.42
3	2.70	3.100	2.720	0.455	0.380	0.75
4	3.60	3.400	3.064	0.244	0.336	0.53
5	3.70	3.900	3.102	0.269	0.798	1.28
6	4.00	2.100	3.216	0.380	-1.116	-1.98
7	3.00	2.800	2.835	0.326	-0.035	-0.06
8	3.30	2.900	2.949	0.238	-0.049	-0.08
9	3.50	3.000	3.025	0.230	-0.025	-0.04
10	4.00	3.800	3.216	0.380	0.584	1.03

Make a write up on the deductions from the output above. (Hint: Your write up should include sections on: (i) Regression equation and the interpretations of coefficients, (ii) R square interpretation, (iii) the hypothesis statement and test conclusions, (iv) correlation value and (v)its interpretation and the fit of the model discussion. **(10 Marks)**

Question 4 (`10 Marks)

(a) During the turn of the millennium, women like Benta (Chief Executive Officer of a manufacturing company) and Sandra (Head of operations at a transport company) had reached the top executive ranks. Yet only some years ago, a surprisingly large percentage of men seemed to believe that woman's place was in the home. In January 2021, Andrew and Pamela published a study estimating the proportion of men in the year 2017, 2018 and 2019 who approved of a married woman earning money (if she has a husband capable of supporting her). The number of men in their sample approved and disapproved each year as follows:

	Year		
	2017	2018	2019
Approve	410	412	409
Disapprove	252	176	151

On the basis of this data, does it appear that men's attitude were changing? Assume that the significance level is 0.05 **(5 Marks)**

(b) An industrial psychologist feels that a big factor in job turnover among assembly line workers is the individual employee's self-esteem. She believes that workers who change jobs often (population A) have, on average, lower self-esteem, as measured by a standardized test, than workers who do not (population B). To determine whether she can support her belief with statistical analysis, she draws a simple random sample of employees from each population and gives a test measuring esteem. The results are summarized in the output below

Descriptive Statistics: Population A, population B

Variable	Total Count	Mean	Standard Dev
Pop. A	10	52.30	9.23
pop. B	17	69.47	11.78

The psychologist believes that the relevant populations of scores are normally distributed, with equal, although unknown variances. At the 0.01 level of significance test whether her belief is valid. Hence make a decision **(5 Marks)**

Question 5 (10 Marks)

A large manufacturing company uses a biometric system at its time clock office. The biometric system identifies the employee and automatically posts the reporting time of an employee to work. Due to its continuous use, the system is inspected for maintenance purposes on a weekly basis. On inspection, the condition of the machine is classified as Good (state 1), minor repairs (state 2), Major repairs (state 3), write off (state 4). The maintenance department adopts the following maintenance policy:

Minor repairs if the system is in state 2, overhaul if the system is in state 3 and replace if the system is in state 4

Policy 1: Minor repairs if the system is in state 1 and 2, overhaul if the system is in state 3 and replace if the system is in state 4.

Suppose the costs of maintainanace associated with each policy are as shown below:

State	Policy 1 (Kshs)	policy 2 (Kshs)
1	0	0
2	50,000	50,000
3	250,000	300,000
4	500,000	500,000

The transition probability Matrices for each policy are given below:

$$\begin{array}{cc} & \text{To} \\ & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \text{From} \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{pmatrix} 0 & 0.8 & 0.1 & 0.1 \\ 0 & 0.5 & 0.4 & 0.1 \\ 0 & 0 & 0.4 & 0.6 \\ 1 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

Determine

(a) the long run proportion of times the system would be expected to be in each state **(6 Marks)**

(b) the policy that would minimize to total cost of maintenance in the long run **(4 Marks)**

Question 6 (10 Marks)

(a) Some managers believe that frequent job changes will lead to rapid advancement than if they remained at the same company. Other managers rarely leave their employers, changing no more than once every 7 years. An article on why and when managers move provided the following data on the average time between moves for 1,191 managers. Assume the average time between moves, X, for each of the 8 time intervals is the midpoint of the interval. These values are shown in the second column table. The percentages shown in the column can be viewed as the approximate probabilities of the respective values of X.

Average time between moves (years)	Mid point, X	Managers who move at this rate,%
1-2	1.5	10.5
2-3	2.5	21.1
3-4	3.5	22.6
4-5	4.5	14.7
5-6	5.5	10.0
6-7	6.5	7.9
7-8	7.5	4.6

8-9	8.5	4.0
9-10	9.5	4.6

(i) Find the expected length of time managers stay on the job **(2 Marks)**

(ii) Find the variance and interpret the result **(3 Marks)**

(b) The academy of Management Journal (August, 2008) did an investigation of the performance and timing of corporate acquisitions. The investigation discovered that in a random sample of 2,778 firms, 748 announced one or more acquisitions during the year 2000. Test whether the sample provide sufficient evidence to indicate that the true percentage of all firms that announce one or more acquisitions during the year 2000 is less than 30%. Use $\alpha = 0.05$ to make your decision. **(5 Marks)**